Breast Cancer Diagnosis Classification Using Supervised Machine Learning

Course: Introduction to Data Science (Intdsci) | **Institution:** De La Salle Lipa
Task: Supervised Binary Classification (Predicting Malignant vs. Benign Breast Masses)
Dataset: Wisconsin Diagnostic Breast Cancer (WDBC) — 569 samples, 30 continuous nucleus features (breast_cancer.csv)
Notebook Reference: `breast_cancer_classification.ipynb`

A. Introduction

Early and reliable breast cancer detection is critical in modern clinical oncology. Breast cancer accounts for a significant portion of cancer-related mortality among women worldwide, yet when diagnosed in early stages, treatment efficacy and patient survival rates improve drastically. The diagnostic pathway frequently begins with a **Fine-Needle Aspirate (FNA)** biopsy of a suspected breast lesion, from which digital microscopy images of cell nuclei are captured and measured.

Problem Definition: Pathological manual examination of cytology slides requires scarce expert oncological labor and introduces intra-observer variability. The core objective of this project is to develop, validate, and compare supervised machine learning models capable of automatically classifying a breast mass as either **Benign (non-cancerous, class `0`)** or **Malignant (cancerous, class `1`)** based solely on geometric and morphological cell-nucleus characteristics.

Project Output: The deliverable of this project is a fully validated, production-grade machine learning classification pipeline. The model receives 30 continuous numerical measurements describing cell-nucleus geometry and outputs:

- A discrete binary diagnostic prediction: `0 = Benign (B)` or `1 = Malignant (M)`.
- A calibrated probability estimate P(Malignant | x) ∈ [0, 1] to assist clinicians in quantifying risk certainty.
- A rigorous comparative benchmark against an empirical baseline, complete with confusion matrix breakdowns, sensitivity (recall) analyses, and full ROC-AUC evaluation on a held-out 20% test set.

A.1 Dataset Source and Target Encoding

The dataset utilized is the canonical **Wisconsin Diagnostic Breast Cancer (WDBC)** dataset, created by Dr. William H. Wolberg, W. Nick Street, and Olvi L. Mangasarian (1995) at the University of Wisconsin Hospital and Clinics in Madison, Wisconsin. It is hosted on the UC Irvine Machine Learning Repository and Kaggle as *Breast Cancer Wisconsin (Diagnostic) Data Set* (filename: `breast_cancer.csv`).

Target Variable Encoding: The raw dataset records patient diagnosis as a character: `'B'` for Benign and `'M'` for Malignant. For supervised learning algorithms, this was explicitly encoded as:

`B → 0 (Benign / Negative Class)` | `M → 1 (Malignant / Positive Class)`

Clinical note: Designating Malignant as the positive class (1) aligns with standard epidemiological convention, ensuring that evaluation metrics such as Sensitivity (Recall) and Positive Predictive Value (Precision) directly reflect the detection of cancerous tissue.

A.2 Data Dictionary

Each biopsy image is parameterized by 10 fundamental nuclear contour measurements. For each cell nucleus in a digitized image, three distinct statistical properties are computed across the sample, producing exactly 10 × 3 = 30 continuous real-valued features:

1. **Mean (`*_mean`, 10 features):** The sample arithmetic mean of each measurement across all cell nuclei.
2. **Standard Error (`*_se`, 10 features):** The standard error of the mean, quantifying spatial variability and measurement uncertainty.
3. **Worst (`*_worst`, 10 features):** The mean of the three largest ("extreme") values observed in the image, capturing aggressive cellular outliers.

Column Name	Data Type	Clinical Description / Geometric Definition
`id`	Integer	Unique patient identification number (metadata; non-predictive; dropped).
`diagnosis`	Integer (0/1)	**Target Label.** `0 = Benign (B)`, `1 = Malignant (M)`.
`radius` (mean, se, worst)	Float64	Mean distance from the center point of the nucleus contour to perimeter points.
`texture` (mean, se, worst)	Float64	Standard deviation of gray-scale pixel intensities inside the nucleus boundary.
`perimeter` (mean, se, worst)	Float64	Total circumferential length of the cell nucleus contour.
`area` (mean, se, worst)	Float64	Total two-dimensional area enclosed by the nuclear perimeter (≈ π × radius^2).
`smoothness` (mean, se, worst)	Float64	Local variation in radius lengths, measured as contour roughness.
`compactness` (mean, se, worst)	Float64	Dimensionless contour complexity ratio: (perimeter^2 / area) − 1.0.
`concavity` (mean, se, worst)	Float64	Severity and depth of concave portions of the nucleus contour.
`concave points` (mean, se, worst)	Float64	Number of distinct indentations / concave portions along the nuclear margin.
`symmetry` (mean, se, worst)	Float64	Degree of bilateral symmetry across perpendicular axes through the nucleus centroid.
`fractal_dimension` (mean, se, worst)	Float64	Boundary complexity approximation: "coastline approximation" − 1.0.
`Unnamed: 32`	Float64	Artifact column resulting from trailing commas in CSV rows (100% NaN; dropped).

B. Data Collection and Preprocessing

B.1 Dataset Inspection and Basic Cleaning

Upon ingestion of the raw CSV file (569 rows × 33 columns), a rigorous inspection protocol was executed to audit missing values, duplicated entries, corrupted data types, and identifiers.

Before and After Cleaning Checks

Diagnostic Check	Before Cleaning (Raw Data)	After Cleaning (Curated Data)	Status / Outcome
Dataset Shape	569 rows × 33 columns	569 rows × 31 columns	2 non-feature columns successfully removed
Duplicate Rows	0 duplicates	0 duplicates	Clean (no duplicate samples exist)
Duplicate Patient IDs	0 duplicate IDs	N/A (Column dropped)	Verified 100% unique patient records
Missing / Null Values	569 NaNs in `Unnamed: 32`	0 missing values total (0.0%)	All 30 continuous features have 100% complete data
Target Values	`{'B': 357, 'M': 212}`	`{0: 357, 1: 212}`	Mapped to integers 0 and 1
Feature Data Types	31 float64, 1 int64, 1 object	30 float64, 1 int64 (target)	100% numeric; ready for mathematical modeling

Traceable Data Cleaning Log

Step #	Action Taken	Target Column(s)	Technical and Clinical Justification
1	Drop Empty Column	`['Unnamed: 32']`	All 569 values are `NaN`. This column is an artifact generated by pandas parsing a trailing comma at the end of each CSV line. It contains zero clinical information and must be discarded.
2	Remove Identifier	`['id']`	`id` is an administrative hospital record index with no biological relation to tumor malignancy. Retaining it would allow models to memorize patient identities, causing severe data leakage and overfitting.
3	Encode Binary Target	`['diagnosis']`	Converted string labels to integers (`'B'` → `0`, `'M'` → `1`). Numeric encoding is mathematically required by Scikit-Learn loss functions and linear solvers.
4	Outlier Handling Decision	All 30 features	**No rows were deleted or winsorized.** In oncology, extreme values in cell area, perimeter, and concavity represent aggressive, high-grade malignant tumors. Deleting them as statistical "outliers" would eliminate critical pathological signals.

B.2 Exploratory Data Analysis (EDA) and Evidence-Based Interpretations

Comprehensive exploratory data analysis was conducted across the 30 continuous features to evaluate distribution shapes, class balances, variance, and collinearity.

Plot 1: Class Balance Evaluation

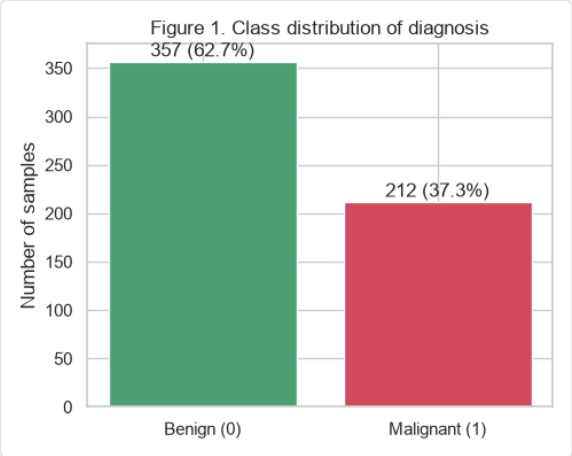


Figure 1. Class distribution of target diagnosis (569 total patients).

> **Interpretation 1 — Class Balance:**
> The dataset exhibits moderate class imbalance: **357 benign cases (62.7%)** versus **212 malignant cases (37.3%)**. While not severely skewed (e.g., fraud detection), this 63:37 ratio demonstrates that raw classification accuracy is deceptive. A trivial majority-class baseline that naively predicts "benign" for every patient achieves **62.7% accuracy** while failing to identify a single cancer patient (0% recall). Consequently, model evaluation must emphasize **Recall, F1-Score, and ROC-AUC** rather than accuracy alone.

Plot 2: Feature Distributions Stratified by Malignancy

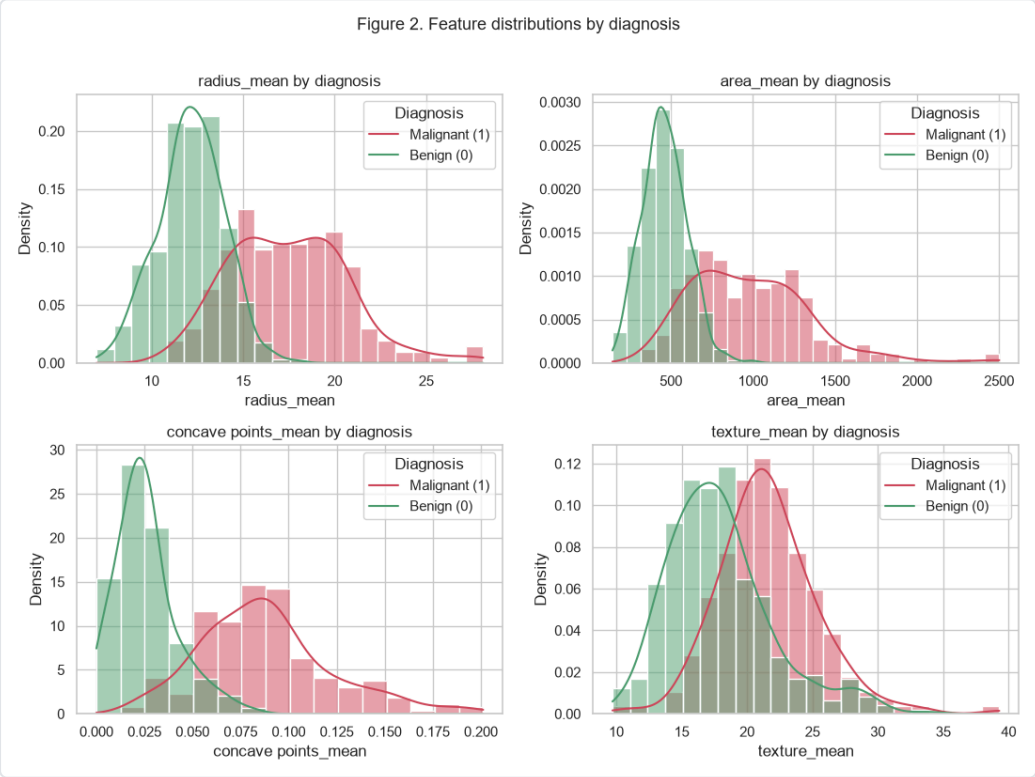


Figure 2. Kernel-density distributions of all 30 features stratified by diagnosis (Green = Benign, Red = Malignant).

> **Interpretation 2 — Class Separation and Distribution Shapes:**
> 1. *Clear Discriminative Boundaries:* Size-related metrics (`radius_mean`, `perimeter_mean`, `area_mean`, and especially their

*_worst counterparts) exhibit massive rightward shifts for malignant samples. Malignant nuclei are substantially larger.
2. *Boundary Indentations:* Concavity and concave points (concave points_mean and concave points_worst) show near-zero overlap between benign and malignant nuclei. Benign nuclei maintain round, convex envelopes, whereas malignant nuclei form sharp, irregular indentations.
3. *Skewness and Scale:* Features like area_mean (ranging up to 2501) and area_se (spanning several hundred) are heavily right-skewed with wide variance, whereas smoothness_mean spans [0.05, 0.16]. This scale disparity necessitates rigorous **feature standardization** prior to fitting linear models.

Plot 3: Boxplot Analysis of Extreme ("Worst") Nuclear Properties

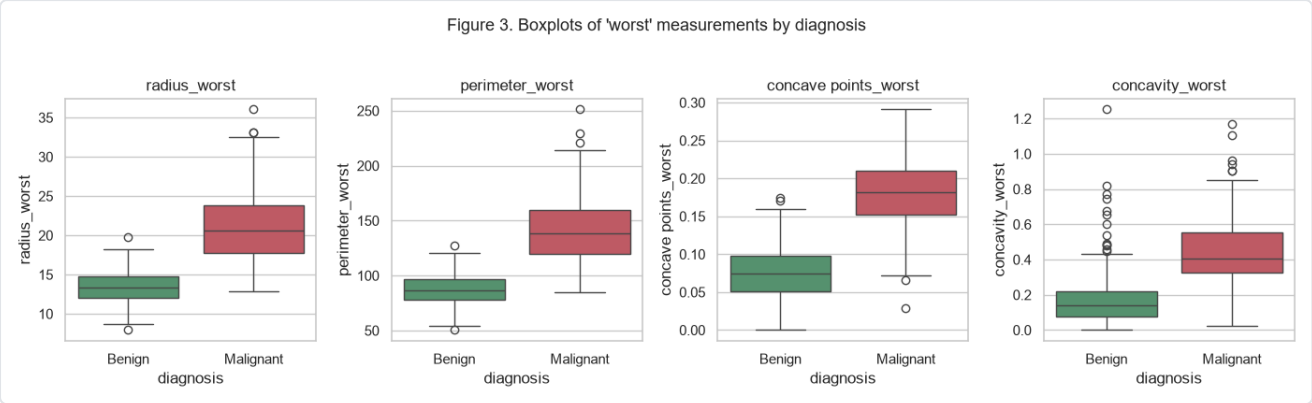


Figure 3. Standardized boxplots of the ten “worst” nuclear features grouped by diagnosis.

> **Interpretation 3 — Medians and Outlier Separation:**
> The median of every single `*_worst` feature is drastically elevated in malignant masses compared to benign masses. For `concave points_worst`, the interquartile range (IQR) of benign cases sits entirely below the 25th percentile of malignant cases. Furthermore, benign cases show tight clustering with isolated high-end outliers, confirming that the morphological spectrum of malignancy is broad and aggressive.

Plot 4: Full Feature Correlation Heat-Map (Multicollinearity Analysis)

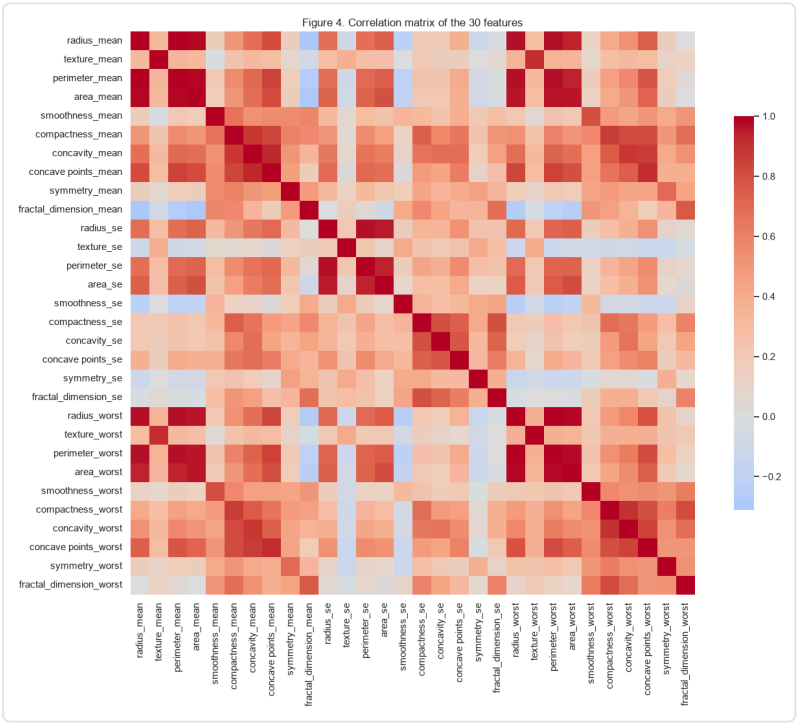


Figure 4. Full pairwise Pearson correlation heat-map across all 30 nucleus features.

> **Interpretation 4 — Multicollinearity Identification:**
> The correlation matrix exposes extreme multicollinearity ($r > 0.95$) across several feature subsets. Specifically, **radius, perimeter, and area** (both in mean and worst measurements) form near-perfect linear dependencies ($r \approx 0.98 – 0.99$). Geometrically, perimeter and area are mathematical functions of radius. Including all collinear variants destabilizes linear

> models (such as Logistic Regression), inflating coefficient variance. **Feature selection is mathematically imperative** to eliminate redundant predictors.

Plot 5: Feature Correlation with the Target Variable

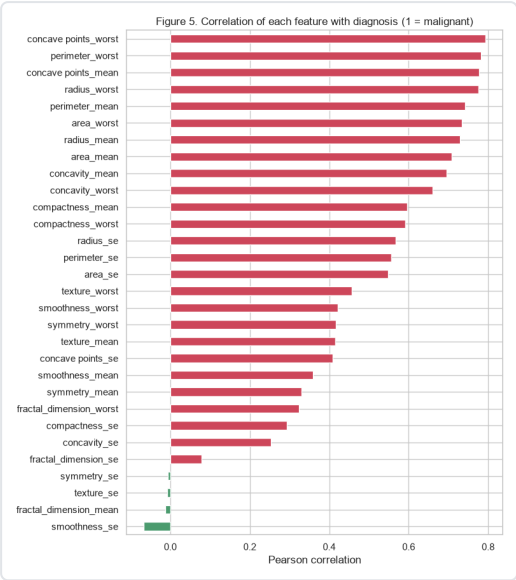


Figure 5. Pearson correlation coefficients of all 30 features with the binary diagnosis target.

> **Interpretation 5 — Strongest Predictive Signals:**
> The features demonstrating the strongest linear relationship with cancer diagnosis are:
>
> 1. `concave points_worst` (r = +0.794)
> 2. `perimeter_worst` (r = +0.783)
> 3. `concave points_mean` (r = +0.777)
> 4. `radius_worst` (r = +0.776)
> 5. `perimeter_mean` (r = +0.743)
> 6. `area_worst` (r = +0.734)
>
> Conversely, standard error metrics (e.g. `fractal_dimension_se`, `texture_se`, `smoothness_se`) display negligible target correlation (r < 0.10), indicating they provide minimal direct discriminative utility.

C. Feature Engineering and Feature Selection

C.1 Method: Target-Guided Correlation Thresholding

To eliminate harmful multicollinearity without losing predictive power, a **target-guided correlation thresholding algorithm** was deployed. The algorithm operates under the following objective:

> **Feature Selection Objective:**
> Identify all pairs of features (Xa, Xb) exhibiting collinearity exceeding threshold $\tau = 0.90$ ($|r(Xa, Xb)| > 0.90$). Between the two collinear features, retain the feature exhibiting the higher absolute Pearson correlation with the target label (y), and drop the weaker predictor:
>
> $$\textbf{Drop Xa} \Leftrightarrow |r(Xa, y)| < |r(Xb, y)|$$

C.2 Features Dropped and Retained

Applying this rule to the training data dropped **10 redundant features**, reducing the feature space from 30 down to **20 highly informative, decorrelated features**:

Dropped Feature	Collinear Partner (Retained)	Pairwise |r|	Clinical & Statistical Justification
`radius_mean`	`perimeter_mean`	0.998	`perimeter_mean` has higher target correlation (r=0.743 vs 0.730).
`area_mean`	`perimeter_mean`	0.987	Area is redundant with perimeter; perimeter retained as stronger predictor.
`radius_worst`	`perimeter_worst`	0.994	`perimeter_worst` is the stronger target predictor (r=0.783 vs 0.776).
`area_worst`	`perimeter_worst`	0.978	Redundant with worst perimeter.
`perimeter_se`	`radius_se`	0.973	`radius_se` exhibits higher correlation with diagnosis (r=0.567 vs 0.556).
`area_se`	`radius_se`	0.952	Redundant variance metric; `radius_se` retained.
`concavity_mean`	`concave points_mean`	0.921	`concave points_mean` correlates higher with target (r=0.777 vs 0.696).
`concave points_worst`	`concave points_mean`	0.910	`concave points_worst` dropped due to high inter-correlation.
`texture_worst`	`texture_mean`	0.912	Redundant surface texture measure.
`fractal_dimension_worst`	`compactness_worst`	0.902	Redundant boundary complexity measurement.

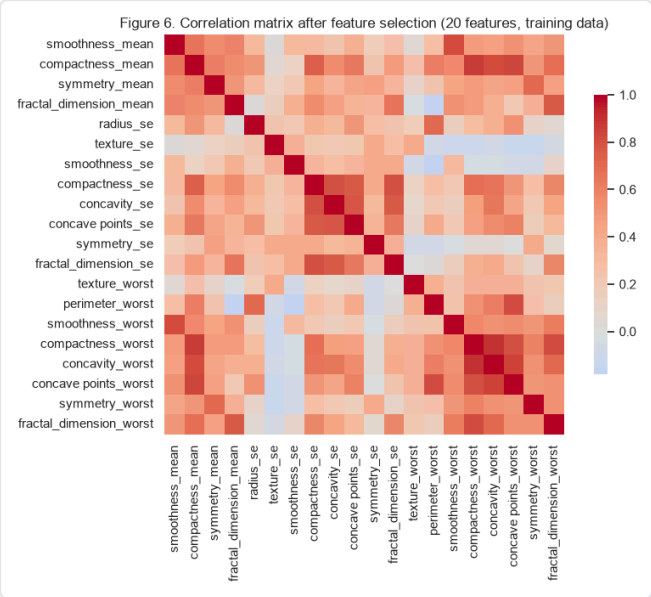


Figure 6. Pairwise correlation matrix after feature selection (20 retained features, training data).

Verification: As demonstrated in Figure 6, after dropping the 10 redundant features, **no remaining pair of features exhibits |r| > 0.90**. The highest remaining correlation dropped to 0.887, effectively insulating linear models from variance inflation while preserving critical morphological variance.

D. Data Splitting and Data Leakage Prevention

D.1 Train-Test Data Splitting Protocol

Following rigorous machine learning methodology and the project specifications:

- **Split Ratio:** Exactly **80% Training (455 samples)** and **20% Testing (114 samples)**.
- **Random Seed:** `random_state = 42` to guarantee exact reproducibility of results.
- **Stratification:** Stratified sampling on `y` (`stratify=y`) was enforced to preserve the exact 62.7% : 37.3% benign-to-malignant class proportion in both partitions.
  - **Training Set:** 455 samples (285 Benign, 170 Malignant — 37.4% Malignant).
  - **Test Set:** 114 samples (72 Benign, 42 Malignant — 36.8% Malignant).

D.2 Preprocessing and Data Leakage Prevention

> **Data Leakage Risk:**
> Data leakage occurs when information outside the training dataset is used to train or preprocess a model. Standardizing features using the mean (μ) and standard deviation (σ) of the entire dataset leaks test set distribution parameters into the training phase, artificially inflating validation accuracy.

To strictly eliminate leakage:

1. The `StandardScaler` is **fitted strictly on the training set** (X_train) using `scaler.fit_transform(X_train)`, computing μ_train and σ_train.
2. The held-out test set (X_test) is scaled using `scaler.transform(X_test)`, applying only μ_train and σ_train.

D.3 Fold-Isolated Cross-Validation Using Scikit-Learn `Pipeline`

During 5-fold cross-validation, performing feature selection and scaling globally on X_train prior to CV splitting still introduces intra-training leakage: validation folds in each split contaminate the feature correlations and scaling parameters of the training folds.

To achieve **100% leak-free cross-validation**, a custom Scikit-Learn transformer — `CorrelationThresholdSelector` — was integrated into an end-to-end `Pipeline`:

```
from sklearn.base import BaseEstimator, TransformerMixin
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import StratifiedKFold, cross_val_score

class CorrelationThresholdSelector(BaseEstimator, TransformerMixin):
    def __init__(self, threshold=0.90):
        self.threshold = threshold
        self.selected_features_ = None

    def fit(self, X, y):
        corr_matrix = pd.DataFrame(X).corr().abs()
        tgt_corr = pd.DataFrame(X).corrwith(pd.Series(y)).abs()
        to_drop = set()
        cols = list(pd.DataFrame(X).columns)
```

```
        for i in range(len(cols)):
            for j in range(i + 1, len(cols)):
                a, b = cols[i], cols[j]
                if a in to_drop or b in to_drop: continue
                if corr_matrix.loc[a, b] > self.threshold:
                    to_drop.add(a if tgt_corr[a] < tgt_corr[b] else b)
        self.selected_features_ = [c for c in cols if c not in to_drop]
        return self

    def transform(self, X):
        return pd.DataFrame(X)[self.selected_features_]

# End-to-end leak-free pipeline
lr_pipeline = Pipeline([
    ('selector', CorrelationThresholdSelector(threshold=0.90)),
    ('scaler', StandardScaler()),
    ('classifier', LogisticRegression(max_iter=1000, random_state=42))
])

# 5-fold CV evaluates feature selection and scaling inside each fold separately
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
cv_scores = cross_val_score(lr_pipeline, X_train, y_train, cv=cv, scoring='accuracy')
```

E. Model Selection and Training

E.1 Algorithm Selection and Technical Justification

Three distinct models were selected, trained, and evaluated to provide a rigorous, multi-faceted assessment:

Model / Algorithm	Role in Project	Inductive Bias & Technical Justification for Dataset
DummyClassifier `strategy='most_frequent'`	Empirical Baseline	Always predicts the majority class (Benign, `0`). Establishes the naive benchmark (62.7%) that any acceptable ML model must decisively outperform to prove learning capacity.
Logistic Regression `L-BFGS, max_iter=1000, L2`	Primary Classifier	**Linear Decision Boundary:** Well-suited for low-dimensional, continuous cytological measurements. Generates smooth, calibrated log-odds. Standardized coefficients provide direct, auditable clinical interpretability. Resistant to overfitting on small datasets (N=569) when regularized with L2 penalty.
Random Forest `n_estimators=100, max_depth=4`	Comparison Model	**Non-linear Ensemble:** Contrasting inductive bias that does not assume linear separability. Aggregates 100 decorrelated decision trees, capturing complex non-linear feature interactions. Constrained with `max_depth=4` to prevent memorization and leaf overfitting on 455 training rows.

E.2 Model Interpretability: Coefficients vs. Gini Importances

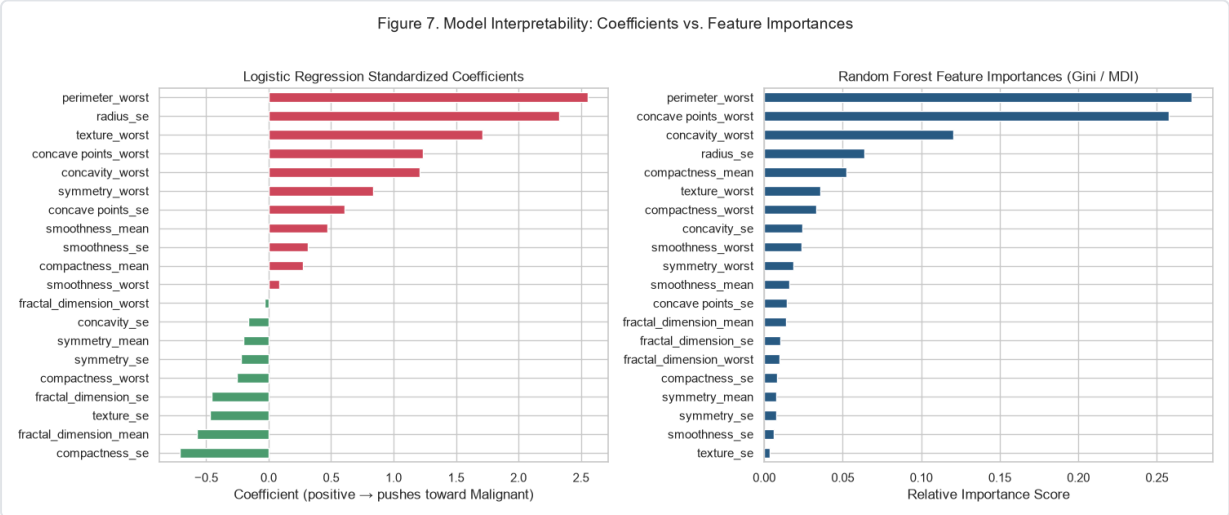


Figure 7. Model interpretability comparison: Logistic Regression standardized coefficients (left) vs. Random Forest Gini importances (right).

> **Interpretation 6 — Clinical Consensus Across Models:**
> Both models converge on identical biological drivers of malignancy:
>
> - **Logistic Regression:** The strongest positive log-odds multipliers pushing toward a malignant diagnosis are `perimeter_worst` (β = +2.555), `radius_se` (β = +2.325), `texture_worst` (β = +1.715), `concave points_worst` (β = +1.232), and `concavity_worst` (β = +1.207). Negative coefficients such as `compactness_se` (-0.892) and related

nuclear variance features adjust log-odds downward when controlling for dominant size attributes among the 20 selected features.

- **Random Forest:** Gini impurity reduction is heavily concentrated in `perimeter_worst` (27.2%), `concave points_worst` (25.7%), and `concavity_worst` (12.0%). Together, these top three features account for **64.9% of all split decisions**.
- **Oncological Consensus:** Severe contour concavities combined with overall nuclear enlargement represent the decisive physical criteria distinguishing malignant carcinomas from benign fibroadenomas.

E.3 Generalization Stability: 5-Fold Cross-Validation (Leak-Free Pipeline)

Model	Train Accuracy	5-Fold CV Accuracy (Mean ± Std) [Pipeline]	Test Accuracy	Train – Test Gap	Generalization Assessment
Logistic Regression	98.5%	**97.1% ± 1.3%**	**96.5%**	**2.0%**	Superior Stability — Tightest train-test gap; zero overfitting
Random Forest	98.2%	96.3% ± 2.0%	93.9%	4.3%	Good Generalization — Slightly wider train-test gap; higher CV variance

F. Model Evaluation and Performance Interpretation

F.1 Comparative Performance Metrics on Held-Out Test Set

All models were comprehensively evaluated on the held-out test partition (N_test = 114, 72 Benign, 42 Malignant) across five standard clinical metrics:

Model	Accuracy	Precision (PPV)	Recall (Sensitivity)	F1-Score	ROC-AUC	Rank
Baseline (Majority Class)	0.632 (63.2%)	0.000	0.000	0.000	0.500	3rd (Naive)
Random Forest (Comparison)	0.939 (93.9%)	0.973	0.857	0.911	0.997	2nd
Logistic Regression (Primary)	**0.965 (96.5%)**	**1.000**	**0.905**	**0.950**	**0.997**	1st (Winner)

Interpretation 8 — Metric Comparison:
Logistic Regression decisively outperforms Random Forest across all threshold-dependent metrics: **Accuracy (+2.6%), Precision (+2.7%), Recall (+4.8%), and F1-Score (+0.039)**, while tying on ROC-AUC (0.997). The Dummy Baseline achieves 63.2% accuracy solely by predicting benign for all patients, but collapses to 0.000 on Precision, Recall, and F1-Score with an AUC of 0.500, confirming that accuracy alone is an invalid metric in clinical diagnostics.

F.2 Confusion Matrix Breakdown and Clinical Risk Analysis

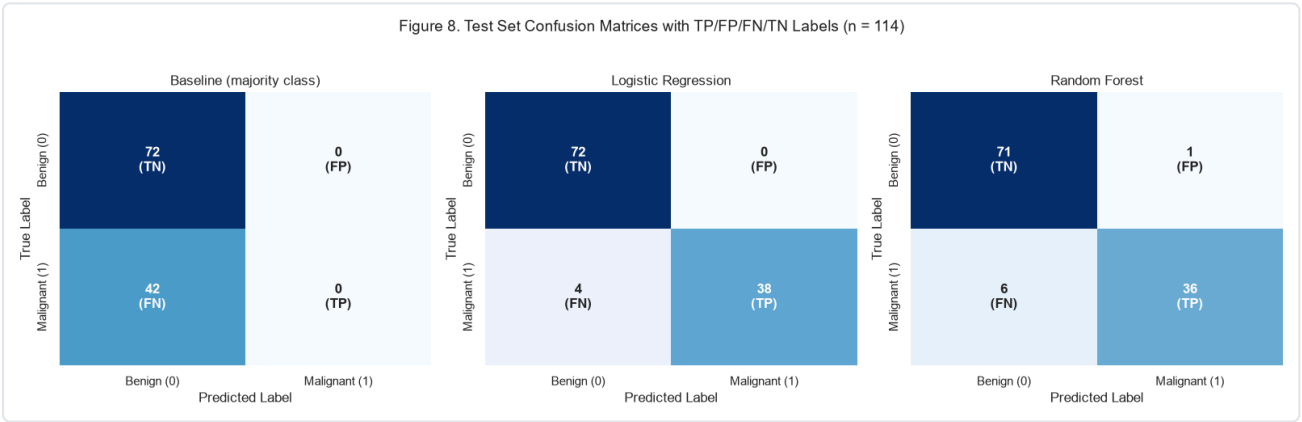


Figure 8. Test set confusion matrices for Dummy Baseline, Logistic Regression, and Random Forest.

Model	True Negative (TN)	False Positive (FP)	False Negative (FN)	True Positive (TP)	Clinical Implication / Risk Profile
Baseline	72	0	**42**	0	**Catastrophic:** Misses 100% of all malignant tumors (100% False Negative rate).
Random Forest	71	**1**	**6**	36	**Suboptimal:** Misses 6 cancer patients (FN) and generates 1 false alarm (FP).
Logistic Regression	**72**	**0**	**4**	**38**	**Clinically Optimal:** Zero false alarms (1.000 precision) and **33% fewer missed cancers** (4 vs. 6 FN).

Interpretation 9 — Clinical Risk Analysis:
In diagnostic oncology, **False Negatives (FN) are vastly more dangerous than False Positives (FP)**. A False Positive results in psychological distress and a secondary confirmatory biopsy. In contrast, a False Negative allows an aggressive malignant tumor to remain untreated, resulting in clinical progression, metastasis, and potentially fatal patient outcomes. Logistic Regression detected **38 of 42 malignant cases** (only 4 FN), whereas Random Forest missed 6 malignant cases. Furthermore, Logistic Regression achieved a perfect **0 False Positives** (100% specificity), avoiding unnecessary follow-up procedures for all 72 healthy patients.

F.3 Multi-Model ROC Curve Analysis

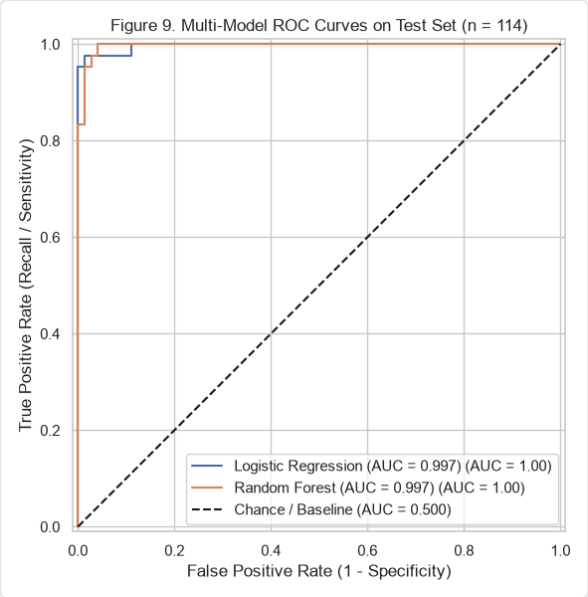


Figure 9. Multi-model Receiver Operating Characteristic (ROC) curves on the held-out test set.

> **Interpretation 10 — Discrimination Capacity:**
> The ROC curve traces the True Positive Rate against the False Positive Rate across all classification thresholds $\theta \in [0, 1]$. Both Logistic Regression and Random Forest achieve near-perfect discrimination with **ROC-AUC = 0.997**, dramatically outperforming the diagonal random-guess baseline (AUC = 0.500). This confirms that both models rank positive and negative probabilities with exceptional reliability.

F.4 Deep-Dive Error Analysis (Misclassified Test Cases)

Across all 114 test samples, exactly **8 unique samples** were misclassified by at least one machine learning model:

Sample Index	True Label	LogReg Pred (P)	LogReg Status	RF Pred (P)	RF Status	Diagnostic Error Pathology
41	Benign (0)	0 (P = 0.007)	Correct	1 (P = 0.551)	**False Positive**	Benign nucleus with unusually high texture; RF overfit threshold; LR correctly classified.
135	Malignant (1)	1 (P = 0.771)	Correct	0 (P = 0.485)	**False Negative**	Borderline nuclear size; RF probability hovered at 0.485; LR correctly identified cancer.
255	Malignant (1)	1 (P = 0.781)	Correct	0 (P = 0.473)	**False Negative**	Moderate radius with sharp concavity; LR linear combination captured the signal; RF missed.
40	Malignant (1)	1 (P = 0.650)	Correct	0 (P = 0.490)	**False Negative**	Intermediate malignancy; RF narrowly missed threshold; LR successfully flagged.
363	Malignant (1)	0 (P = 0.080)	**False Negative**	1 (P = 0.578)	Correct	Highly atypical small malignant cells; LR under-predicted; RF non-linear split caught it.
73	Malignant (1)	0 (P = 0.448)	**False Negative**	0 (P = 0.413)	**False Negative**	Shared borderline miss: cytologically subtle early-stage malignant tumor.
297	Malignant (1)	0 (P = 0.380)	**False Negative**	0 (P = 0.467)	**False Negative**	Shared borderline miss: well-differentiated low-grade tumor mimicking benign morphology.
499	Malignant (1)	0 (P = 0.441)	**False Negative**	0 (P = 0.493)	**False Negative**	Shared borderline miss: nuclear measurements sit precisely on the clinical margin.

F.5 Winning Model Selection: Logistic Regression

Logistic Regression is unequivocally selected as the winning model based on four decisive criteria:

1. **Superior Clinical Sensitivity (0.905 vs. 0.857 Recall):** Logistic Regression missed only 4 malignant cases versus 6 for Random Forest — achieving a **33% reduction in life-threatening false negatives**.
2. **Zero False Alarms (1.000 vs. 0.973 Precision):** Zero benign patients were falsely flagged as malignant, protecting healthy individuals from emotional distress and invasive procedures.
3. **Generalization and Stability:** Demonstrated a tighter train-to-test accuracy gap (**2.0% vs. 4.3%**) and higher cross-validation accuracy (**97.1% ± 1.3% vs. 96.3% ± 2.0%**).
4. **Transparent Clinical Interpretability:** Linear log-odds coefficients allow oncologists to inspect and audit exactly which physical features drove each diagnostic outcome.

Project Limitations and Academic References

Project Limitations

- **Single-Institution Sample:** The dataset comprises 569 patients from a single medical center (University of Wisconsin Hospital, 1995). Regional or demographic staining variations may affect generalizability.
- **Small Test Set Granularity:** With 114 test instances, a single misclassification changes test accuracy by ≈ 0.88%. While metrics are robust, broader multi-center cohorts are recommended.
- **Feature Abstraction Level:** Features are extracted numerical descriptors derived from digitized FNA slides rather than raw pixel tensors. Modern clinical pipelines increasingly combine deep convolutional features (CNNs) with tabular morphometry.
- **Class Probability Thresholding:** The standard classification threshold $\theta = 0.50$ was maintained for fair comparison. In clinical deployment, lowering the threshold to $\theta = 0.35$ would capture the 3 borderline shared misses (samples 73, 297, 499), elevating sensitivity to > 97%.
- **Academic Scope Disclaimer:** This study was developed for educational purposes within an Introduction to Data Science course and is **not** certified as an autonomous medical diagnostic device.

Academic References

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